TG-40 (Daily QA)

Wednesday, July 31, 2024 7:46 AM

TG40 TG142 TG198...

Daily QA:

- Tests that can seriously affect treatment accuracy
    - Patient position
    - Patient dose
    - Safety
- Include parameters that can affect dose delivery by
    - Dosimetric (output constancy)
    - Geometric (laser, ODI, field size)
- Performed typically by morning warmup therapist
    - Trained by QMP with well-defined policy and procedure

Table I. Daily.

Procedure	Machine-type tolerance Non-IMRT	IMRT	SRS/SBRT	TG-40	
Dosimetry					
X-ray output constancy (all energies) Electron output constancy (weekly, except for machines with unique e-monitoring requiring daily)		3%			Same
Mechanical					
Laser localization	2 mm	1.5 mm	1 mm	2 mm	2 mm
Distance indicator (ODI) @ iso	2 mm	2 mm	2 mm		Same
Collimator size indicator	2 mm	2 mm	1 mm	None	2 mm
Safety					
Door interlock (beam off)		Functional			Same
Door closing safety		Functional		None	
Audiovisual monitor(s)		Functional			Same
Stereotactic interlocks (lockout)	NA	NA	Functional		Same
Radiation area monitor (if used)		Functional			
Beam on indicator		Functional		None	Same

- **Daily Imaging**
    - Trained by QMP with well-defined policy and procedure

	Daily[a]	
Planar kV and MV (EPID) imaging		
Collision interlocks	Functional	Functional
Positioning/repositioning	≤ 2 mm	≤ 1 mm
Imaging and treatment coordinate coincidence (single gantry angle)	≤ 2 mm	≤ 1 mm
Cone-beam CT (kV and MV)		
Collision interlocks	Functional	Functional
Imaging and treatment coordinate coincidence	≤ 2 mm	≤ 1 mm
Positioning/repositioning	≤ 1 mm	≤ 1 mm

- **Daily Equipment**
    - Daily QA 3 by Sun Nuclear
        - 13 ion chambers
            - 5 ion chambers for flatness/symmetry
            - 4 ion chambers for electron energy
            - 4 ion chambers for photon energy
        - 12 diodes
            - Light-radiation coincidence
    - Isocube
        - Central BB, offset fiducials
            - Laser localization
            - Positioning/repositioning

★ Annual QA:

Subset of the tests performed during acceptance and commissioning

Baselines established, reconfirmed, or updated

Performed by QMP

PCAM standards used:

- Published department data
- AAPM TG-40/142
- Original Acceptance Test Data
- Varian Acceptance Test Specifications
- PA State Regulations

Annual Mechanical/Safety:

TABLE III. Annual.

	Machine-type tolerance				
Procedure	Non-IMRT	IMRT	SRS/SBRT	TG-40	
Mechanical					
Collimator rotation isocenter		±1 mm from baseline			Same
Gantry rotation isocenter		±1 mm from baseline			Same
Couch rotation isocenter		±1 mm from baseline		2 mm diameter	Same
Electron applicator interlocks		Functional			Same
Coincidence of radiation and mechanical isocenter	±2 mm from baseline	±2 mm from baseline	±1 mm from baseline		2 mm
Table top sag		2 mm from baseline			
Table angle		1°		None	Same
Table travel maximum range movement in all directions		±2 mm		Only vertical	Same
Stereotactic accessories, lockouts, etc.	NA	NA	Functional	None	Same
Safety					
Follow manufacturer's test procedures		Functional			Same
Respiratory gating					
Beam energy constancy		2%			
Temporal accuracy of phase/amplitude gate on		100 ms of expected		None	N/A
Calibration of surrogate for respiratory phase/amplitude		100 ms of expected			
Interlock testing		Functional			

Annual Dosimetry:

TABLE III. Annual.

	Machine-type tolerance				
Procedure	Non-IMRT	IMRT	SRS/SBRT	TG-40	
Dosimetry					
X-ray flatness change from baseline		1%			1%
X-ray symmetry change from baseline		±1%		N/A	2% Abs
Electron flatness change from baseline		1%			2%
Electron symmetry change from baseline		±1%			3% Abs
SRS arc rotation mode (range: 0.5–10 MU/deg)	NA	NA	Monitor units set vs delivered: 1.0 MU or 2% (whichever is greater) Gantry arc set vs delivered: 1.0° or 2% (whichever is greater)		
X-ray/electron output calibration (TG-51)		±1% (absolute)		2%	2%
Spot check of field size dependent output factors for x ray (two or more FSs)		2% for field size $<4\times4$ cm^2, 1% $\geq 4\times4$ cm^2		2%	1%
Output factors for electron applicators (spot check of one applicator/energy)		±2% from baseline			2%
X-ray beam quality (PDD_{10} or TMR^{20}_{10})		±1% from baseline			PDD10 1%, other 2%
Electron beam quality (R_{50})		±1 mm			R50 1mm, other 2mm
Physical wedge transmission		±2%			None

factor constancy					
X-ray monitor unit linearity (output constancy)	±2% ≥5 MU	±5% (2–4 MU), ±2% ≥5 MU	±5% (2–4 MU), ±2% ≥5 MU	1%	1%
Electron monitor unit linearity (output constancy)		±2% ≥5 MU			
X-ray output constancy vs dose rate		±2% from baseline		N/A	2%
X-ray output constancy vs gantry angle		±1% from baseline		2%	1%
Electron output constancy vs gantry angle		±1% from baseline		2%	1%
Electron and x-ray off-axis factor constancy vs gantry angle		±1% from baseline		2%	1%
Arc mode (expected MU, degrees)		±1% from baseline			
TBI/TSET mode		Functional		None	Same
PDD or TMR and OAF constancy		1% (TBI) or 1 mm PDD shift (TSET) from baseline			Same
TBI/TSET output calibration		2% from baseline			Same
TBI/TSET accessories		2% from baseline			Same

Annual Imaging:

TABLE VI. Imaging.

Procedure	Application-type tolerance: non-SRS/SBRT	Application-type tolerance: SRS/SBRT	
Annual (A)			
Planar MV imaging (EPID)			
Full range of travel SDD	±5 mm	±5 mm	Radiation Safety
Imaging dose[e]	Baseline	Baseline	
Planar kV imaging			
Beam quality/energy	Baseline	Baseline	
Imaging dose	Baseline	Baseline	
Cone-beam CT (kV and MV)			
Imaging dose	Baseline	Baseline	

*Annual review of imaging dose deemed sufficient due to minimal effect on overall dose and existing daily/monthly reviews of parameters that could affect dose

**UPenn Radiation Safety responsible for tube and exposure calibration checks

Annual VMAT guidelines (TG-218):

- Ensures linac can deliver VMAT plans that pass institutional VMAT QA criteria based on TG218 guidelines.
- Tests accuracy of VMAT delivery when treatment is interrupted and restarted

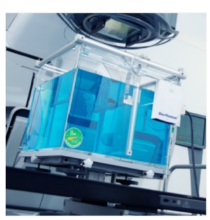
VMAT				
Test patient-specific VMAT QA constancy	Plan passes QA at a pass rate ≥ baseline pass rate minus 2% AND Plan passes institutional VMAT QA criteria	VMAT QA phantom	20–30 min	QMP
Interruption test	± 1% deviation from uninterrupted passing rate	VMAT QA phantom	10–15 min	QMP

Annual Equipment:

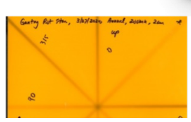
- Blue Phantom 2 by IBA Dosimetry
  - Large 3D water tank with robotic arms

Water Tank

Exterior water tank dimensions (LxWxH):	675 mm x 645 mm x 560 mm
Scanning volume (LxWxH):	480 mm x 480 mm x 410 mm
Position resolution:	0.1 mm
Position accuracy:	± 0.1 mm
Position reproducibility:	± 0.1 mm
Scanning speed:	max. 50 mm/s
Approximate volume:	200 l
Wall thickness / material:	15 mm / acrylic
Weight:	45 kg



- 1D Scanner Water Tank by Sun Nuclear
  - Smaller 1D scanner for output factors/TG-51



- Radiochromic Film
  - Star Patterns to test isocentricity